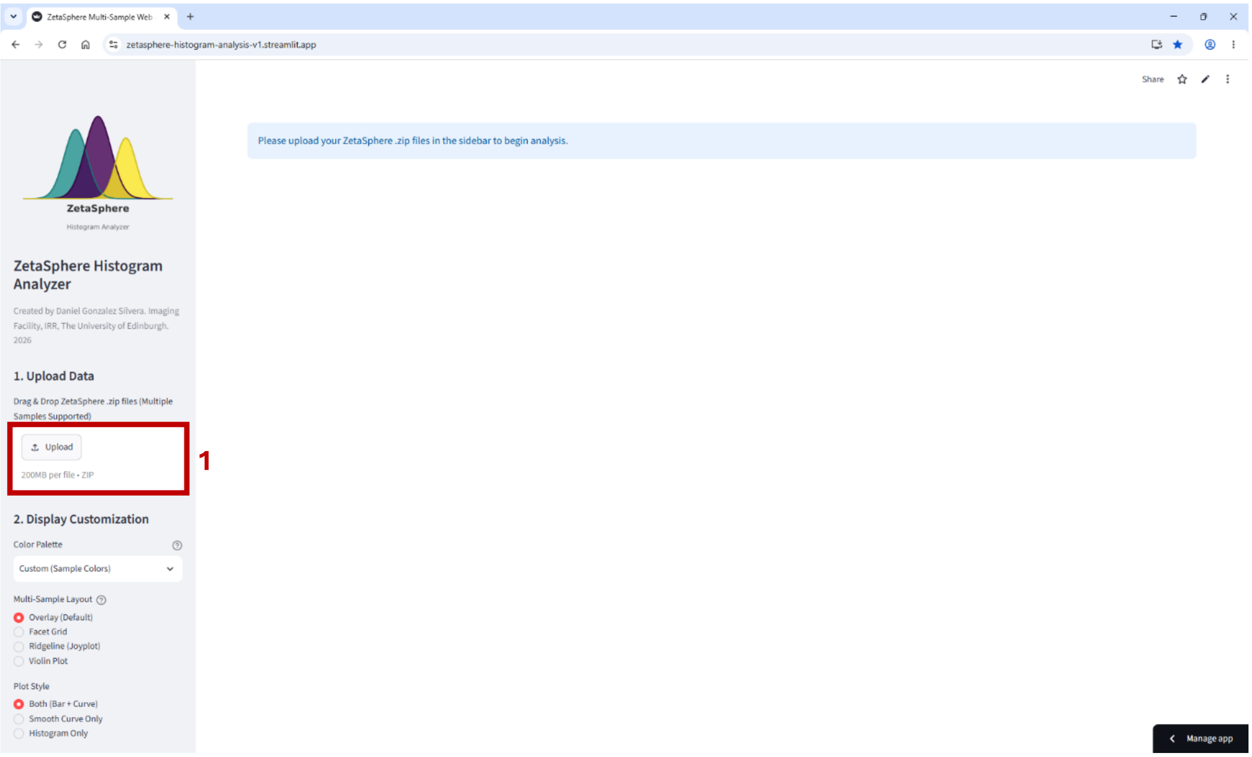
Nanoparticle Histogram Analyzer app

Custom software for easy generation and export of publication-quality histograms and graphs for size, zeta potential, concentration and colocalization results, including analysis of histograms to detect sub-populations, automatic statistics and colocalization quality controls.

Loading app an uplaoding files

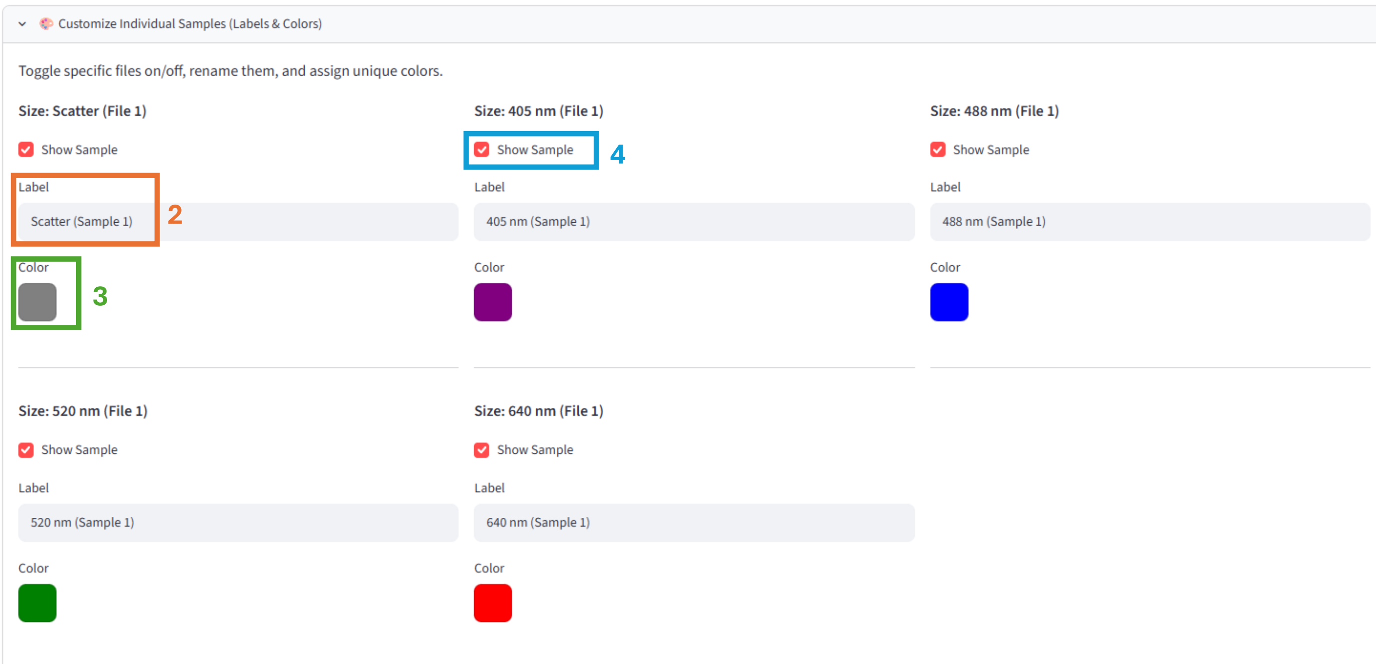
You can access from any computer on https://nanoparticle-histogram-analyzer.streamlit.app/

A web browser window will pop up:

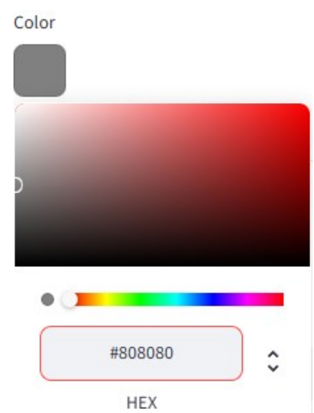


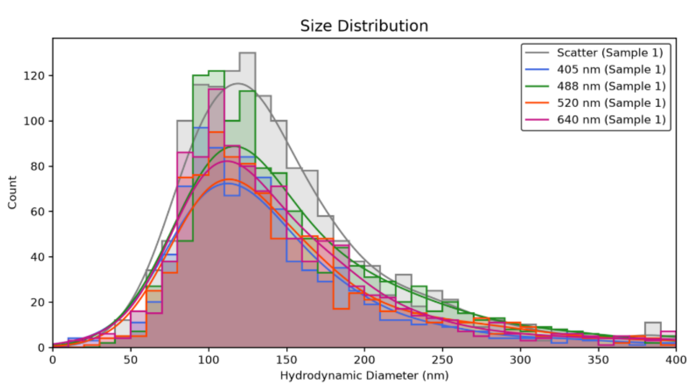
You can upload in **(1)** the zip files containing the raw data for any kind of measurement. The software automatically detects the type of measurement and channel used from the name of the file. You can add different measurement/channels of the same sample, but also files from different samples and pool or compare them.

All uploaded files will be shown in the "Customize individual Samples" section. Here, you can change the name of the sample (2), the colour to represent it (3), and also you can untick (4) any sample to remove it from the graphs.



If you upload measurements from the same sample in scatter and fluorescence mode, the app automatically assigns grey colour for scatter and a representative laser excitation colour for each fluorescence channel. You can modify the default colour by clicking in the coloured squares (3) and select a colour from the palette or type its hexadecimal code.



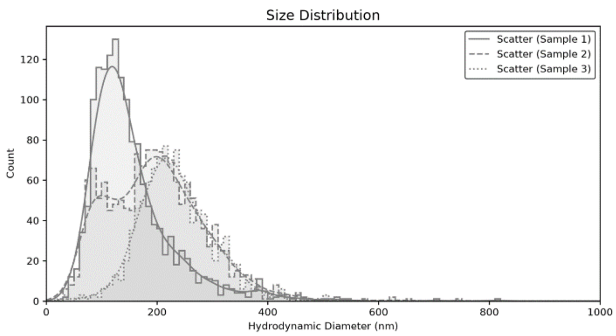


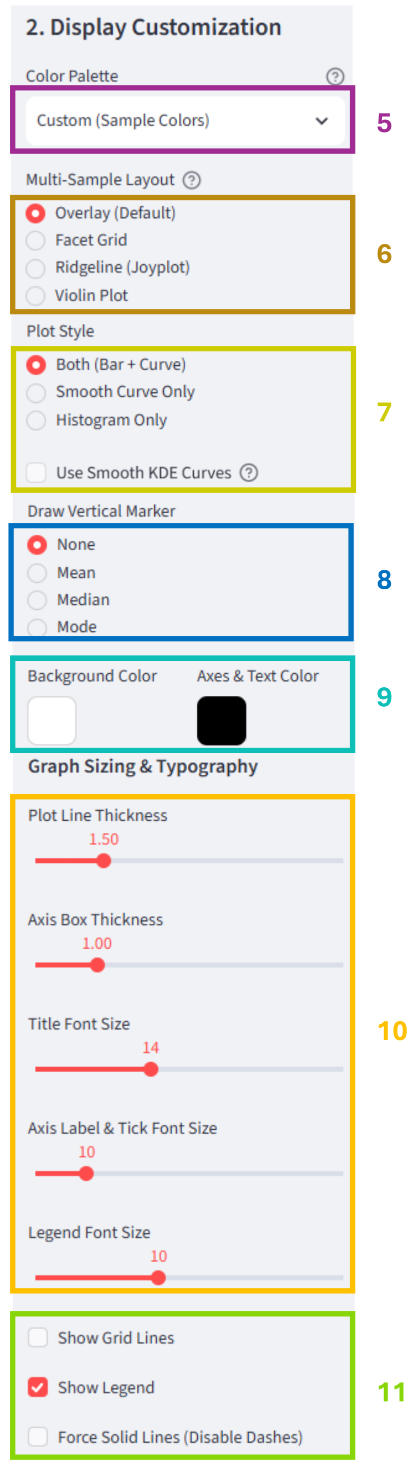
Example of a default size graph from a sample in scatter mode and all fluorescence channels.

Display Customization section

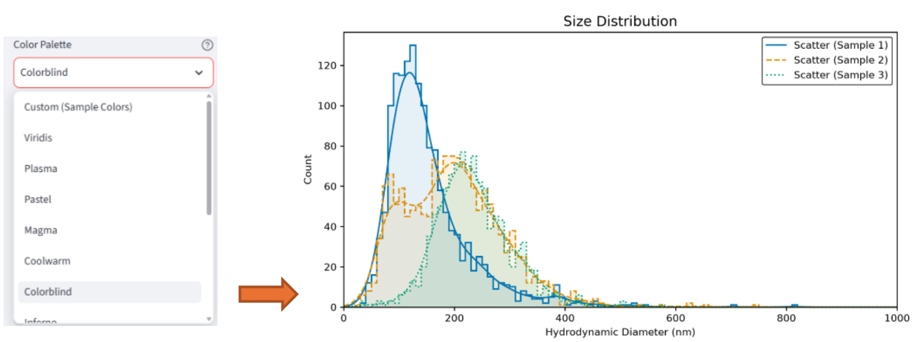
We will use as an example the size results from three different samples.

At the left-hand side of the window, below the “upload data” section, there are different sliders and options to customize your graph.

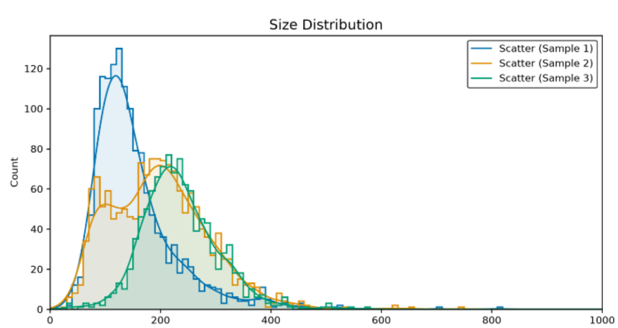




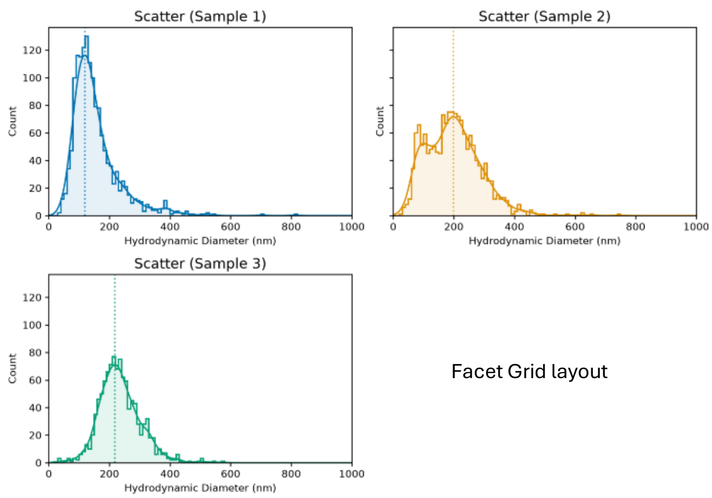
By default, different samples in scatter mode are shown in grey and differentiated by dashed lines. You can change the colours individually as explained above, but you can also apply predefined colour palettes. Just click on **(5)** and select a palette from the list. The graph will be automatically updated:

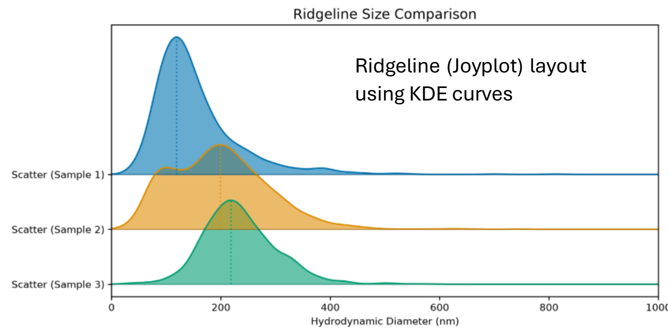


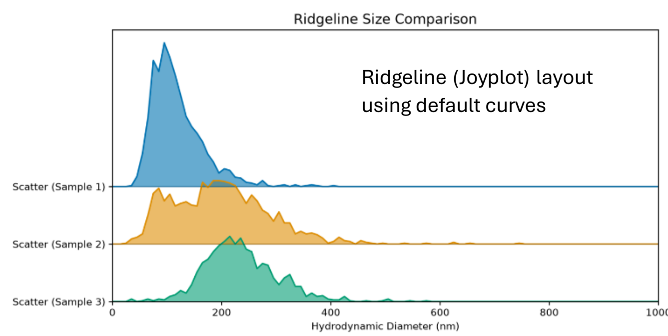
You can remove the dashed pattern by clicking on “Force Solid lines” **(11)**:

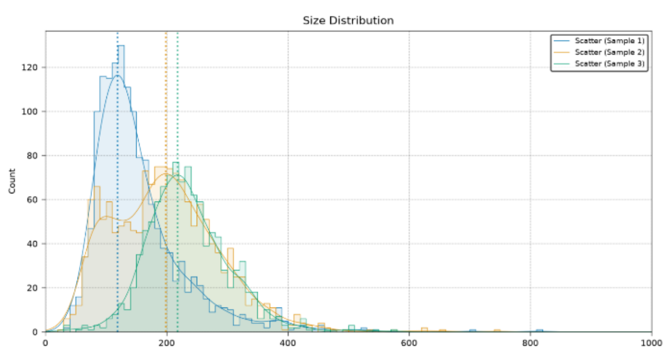


Click on “Multi-Sample Layout” **(6)** to modify how the different samples are displayed: Overlay (default option), Grid, Ridgeline and Violin plot. Click on “Plot Style” **(7)** to only show the histogram, the smooth curve, or both (default). The smooth curve is drawn based on the middle points of each bin, and therefore its shape depends on the bin size. Clicking on “Use smooth KDE curves” will show a smoother curve, drawn based on Kernel Density Estimation and therefore independent of the bin size. Clicking on “Draw Vertical Marker” **(8)** allows you to visualize the position of the Mean, Median or Mode. Click on **(9)** to modify the colour of the background and the text/axes.







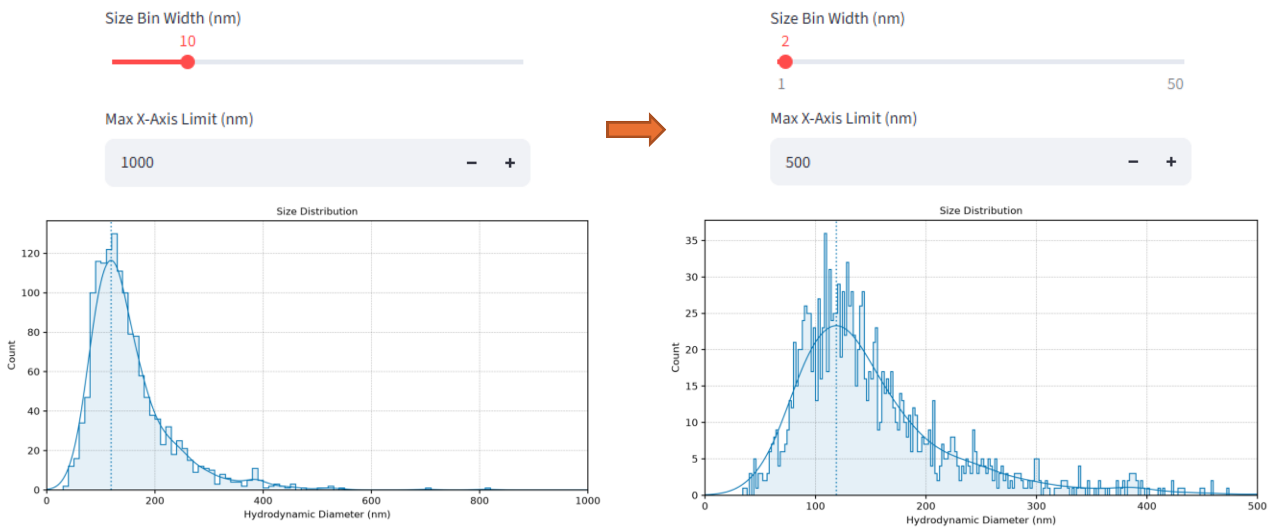


The “Graph Sizing and Typography” section **(10)** contains different sliders to control the thickness of the plot lines, the thickness of the axis, the font size of the title, the font size of axis labels and tick, and the font size of the legend. You can show/hide the legend and grid lines in **(11)**. You can also select the legend position.

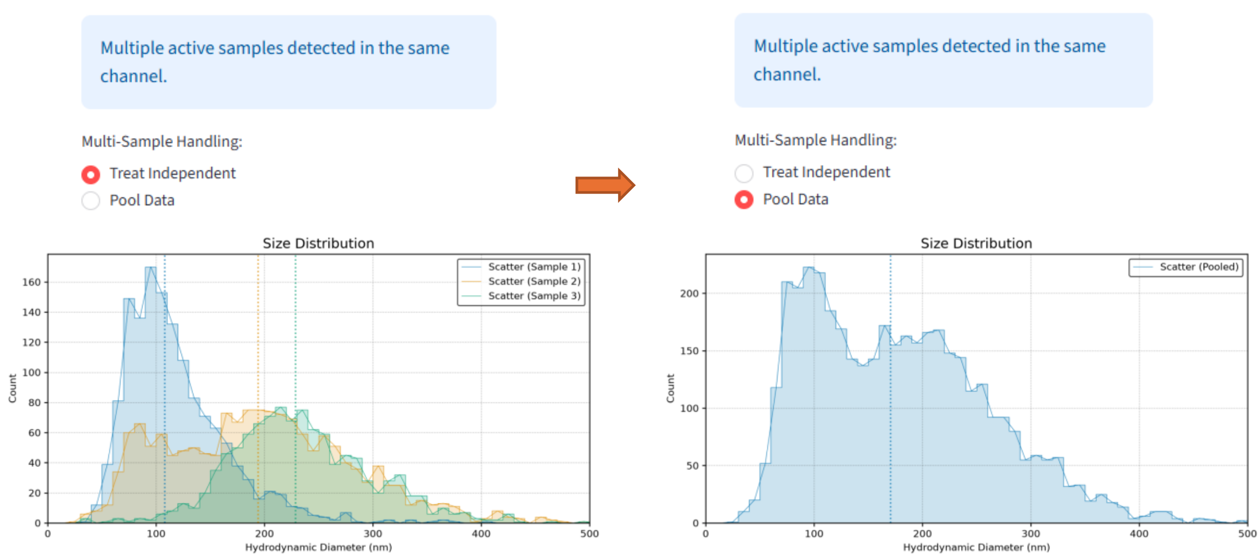
There are 6 different tabs below the “Customize Individual Samples” section, one for each type of measurement (Size, Zeta potential, concentration and Colocalization), one for subpopulations analysis in a histogram (Population Analysis) and one for extra measurements from a colocalization result (Colocalization Quality). **All the graphs can be downloaded in .png and .svg fomats**.

Multi-Sample Size tab

Select this tab to visualize your particle size distributions. In addition to the graph customization, you can modify the bin size and the limit for the x-axis:



If the samples are actually different replicates, there is a button that will pool all of them and create a histogram with the sum of all individual particles from all active samples.



Below the graph, the app shows a table with different data, including the count, mean, median, mode, standard deviation (SD), median absolute deviation (MAD), span and full width at half maximum (FWHM). Below this table there is a second table with statistics among all active samples.

📊 Statistical Comparison (Size)

	Sample	Count	Mean (nm)	Median (nm)	Mode (nm)	SD (nm)	MAD (nm)	Span	FWHM (nm)
0	Scatter (Sample 1)	1485	153.7	133.5	119.1	77.2	34.9	1.232	104.2
1	Scatter (Sample 2)	1546	197.8	194	198.8	88.7	61	1.158	223.6
2	Scatter (Sample 3)	1059	235.1	228.5	218.4	67.4	40.5	0.721	130.5

📚 **Understanding the Statistical Tests:**

- **Mann-Whitney U:** Gives a p-value for the shift in the median.
- **Anderson-Darling:** Gives a p-value testing if the overall shape and tails of the distributions are different.
- **Earth Mover's Distance (EMD):** This does not yield a p-value. Instead, it gives a physical "Distance Score" (in nm or mV). A score of 0.0 means the histograms are identical. The higher the number, the more physical "work" is required to make one curve look like the other.

All-vs-All Pairwise Comparisons

	Sample A	Sample B	MW p-value	EMD Score	AD p-value
0	Scatter (Sample 1)	Scatter (Sample 2)	< 0.001	44.42	< 0.001
1	Scatter (Sample 1)	Scatter (Sample 3)	< 0.001	82.08	< 0.001
2	Scatter (Sample 2)	Scatter (Sample 3)	< 0.001	38.50	< 0.001

It is **not recommended** to use the **mean** in nanoparticle size analysis, as the histograms are always skewed to right (due to clusters or presence of big artifacts). Use the **median** or **mode** (most frequent particle size) instead. The mean and median values given by this app are the same than the ones provided by ZetaSphere, but not the mode. This app calculates the mode using a high resolution Gausian Kernel Density Estimation, while ZetaSphere may use a frequency polygon.

Definition of terms and statistics

The **mean** is the arithmetic average particle size of the particle size distribution.

The **median** is the particle size at which 50% of the distribution is smaller.

The **mode** is the most frequently occurring particle size in the distribution (peak of the

distribution).

The **SD** (Standard Deviation) indicates the spread of particle sizes around the mean.

The **MAD** (Median Absolute Deviation) is a highly resilient measure of statistical dispersion that is less sensitive to extreme outliers (aggregates) than Standard Deviation. Very good to measure population width (homogeneity of the nanoparticle population). *Note: If your SD is much larger than MAD, the sample may have some big aggregates skewing the average*.

The **span** is another measure of population width, calculated using percentiles. Lower span indicates higher uniform, size distribution. It is less robust than MAD, but allows you to compare with other devices measuring PDI (polydispersity index).

The **FWHM** (nm) is best for measuring homogeneity of individual populations (while MAD and span describe the entire histogram). It measures how wide is a peak at 50% of its height, ignoring the rest of the histogram.

The **bin size** determines how big are the histogram bars that group particles of that specific range of sizes (or mV in zeta potential histograms). Small bins give a more detailed histogram, but very noisy and spikey. Large bins give a smooth and generalized shape, but with less resolution and with the risk of hiding different populations.

The **KDE (Kernel Density Estimation)** is the smooth curve that you can visualize as an alternative or complementary to the histogram bins. KDE is mathematically continuous and doesn't rely on rigid bin sizes, therefore it is accurate to find the true centre peak (the mode) of sub-populations, completely independent of how the histogram bins are defined.

The **hydrodynamic diameter** is what you are really measuring when you refer to a Zetaview size measurement. ZetaSphere tracks the Brownian movement of particles. Small particles move quickly, big particles move slowly. The software calculates the size of a perfect sphere that would move at that same speed.

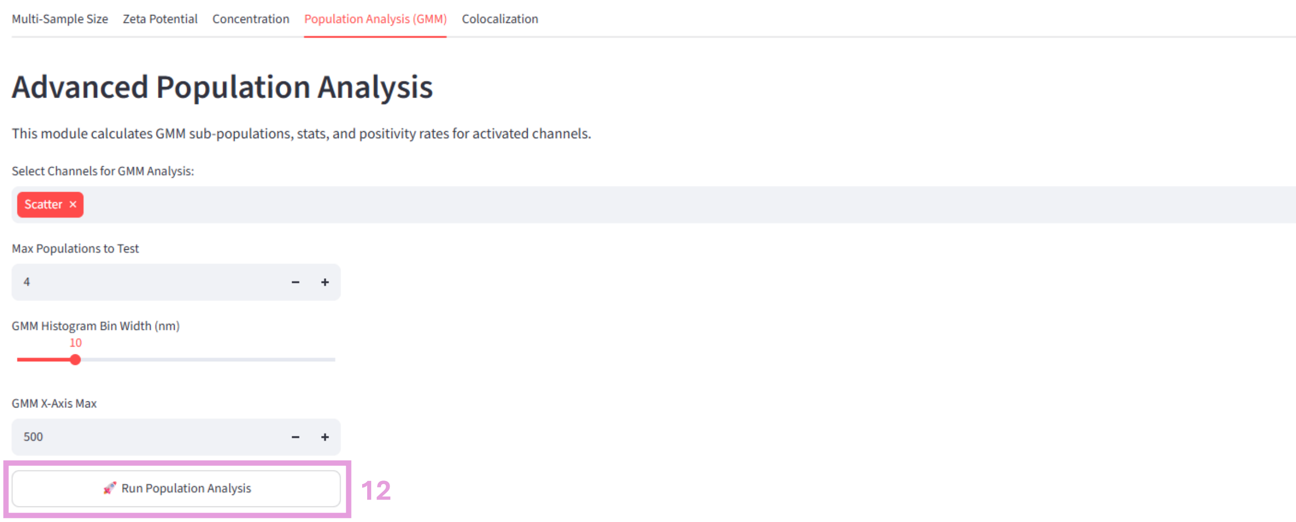
Statistics: **Mann-Whitney U test (MW p-value)**. Gives a p-value for the shift in the *median*. While the Mann-Whitney U test is great for finding if one median is larger than another, it completely ignores the *shape* of the histogram (for example, height of the peak, or presence of different subpopulations).

Statistics: **Earth Mover's Distance (EMD score)**. Instead of a p-value, it gives a physical "distance score" (in nm or mV). A score of 0.0 means the histograms are identical. The higher the number, the more physical "work" is required to make one curve look like the other. It calculates the "cost" of physically moving the particles of Histogram A to make it look exactly like Histogram B.

Statistics: **Anderson-Darling k-sample test (AD p-value)**. It tests the null hypothesis that your samples are drawn from the exact same underlying population. Gives a p-value testing if the overall *shape and tails* of the distributions are different.

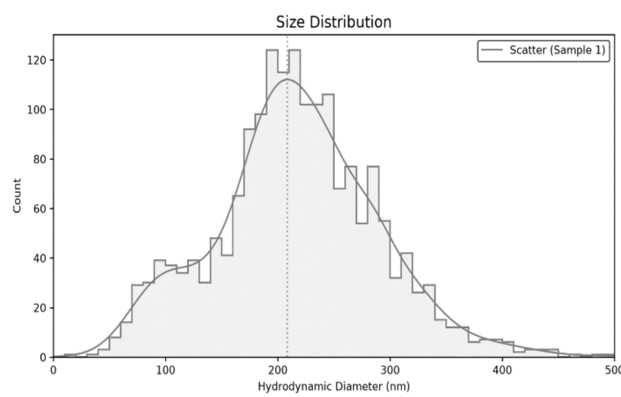
Population Analysis (GMM) tab

This module utilizes Gaussian Mixture Models (GMM) to mathematically separate overlapping nanoparticle size distributions into distinct sub-populations.

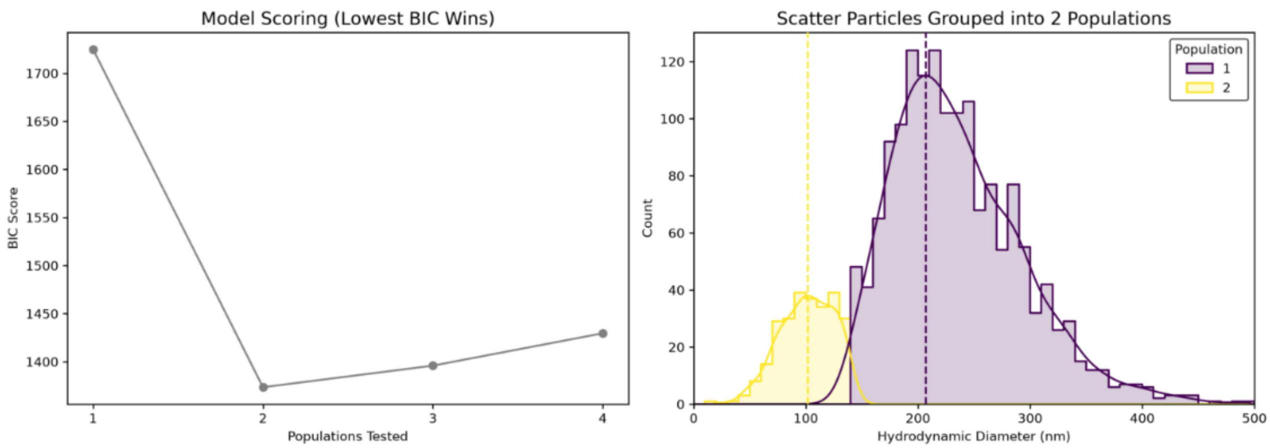


Once your size samples are uploaded, you can select the maximum number of populations to test, the bin size and the maximum X-axis value (most of the options on the “Display Customizations” section still apply here). Just click on (12) Run Population Analysis button and the app will analyse the histogram.

This example shows the size histogram of a mixture of 100nm and 200nm beads. The mode marks the most abundant size in the sample, which is close to 200nm and does not detect the smaller, less abundant population.



After a few seconds, the app will show a series of tables with the data and a combined graph:



The left-hand side graph shows the BIC score pointing to two populations in the sample (see next paragraph for an explanation) and the right-hand side graph shows the size histogram with both populations separated, each of them with their own vertical mark pointing to their specific mode values.

To determine the number of populations, the app uses a statistical scoring method called the **Bayesian Information Criterion (BIC)**. The left-side graph shows the BIC score for each population model tested. The app tests how accurate is to separate the histogram in different populations and the number of populations with the lowest score shows the highest probability of being real. The right-side graph shows the size histogram, separating the populations using different colours (customizable with the colour palette option).

The app will generate above the graph two tables, one with general information and a second one with sub-population details.

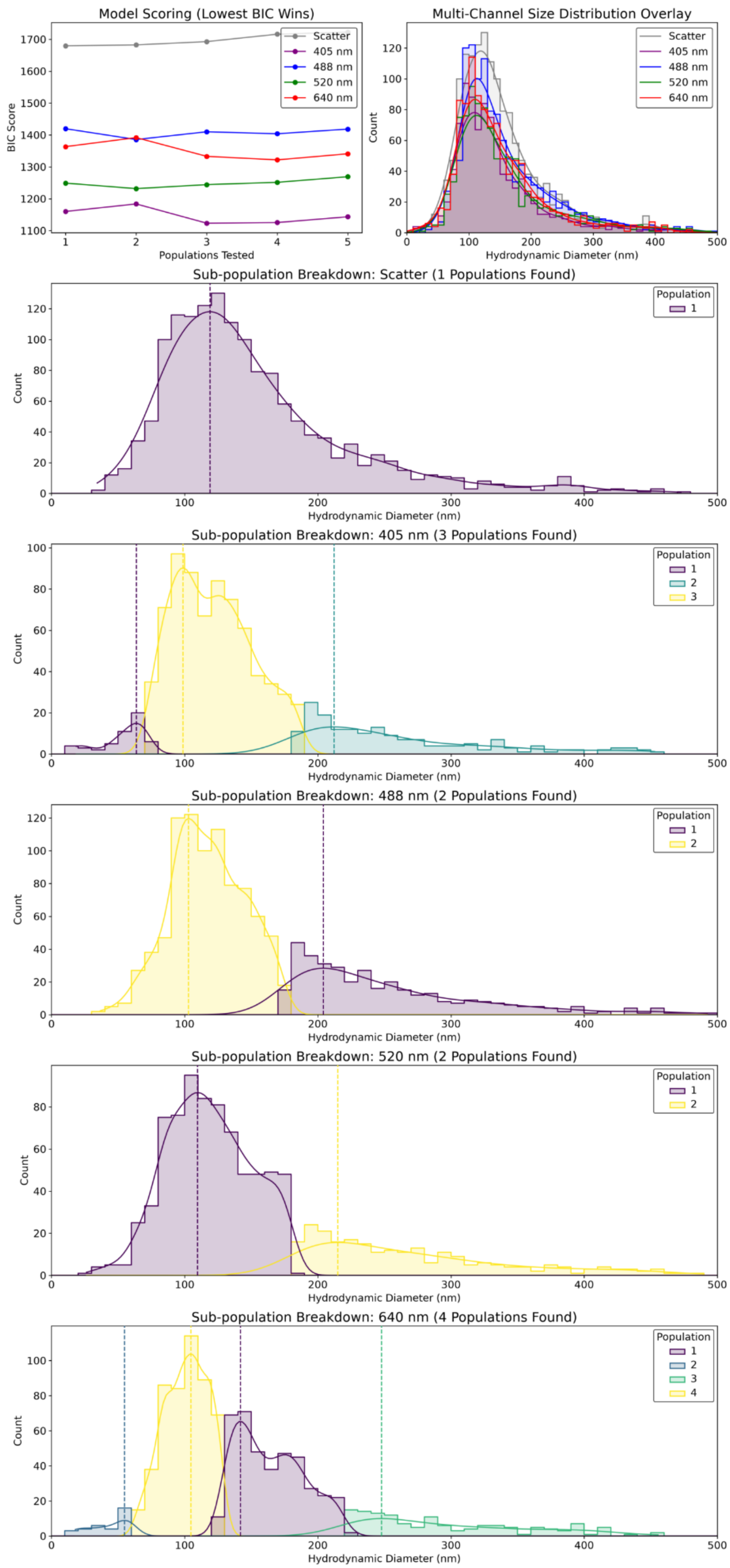
Global Overview

	Channel	Total Particle Count	Global Median Size (nm)	Positivity Rate
0	Scatter	1821	214.76	N/A

Sub-Population Details

Scatter Channel Breakdown:

	Channel	Population	Count	Span	Percentage (%)	Mean (nm)	Median (nm)	Mode (nm)	SD (nm)	MAD (nm)	FWHM (nm)
0	Scatter	1	1556	0.634	85.4476	236.3	227.1	206.9	57.9	37.6	131.1
1	Scatter	2	265	0.597	14.5524	101	102.5	101.8	24	18.2	70.3



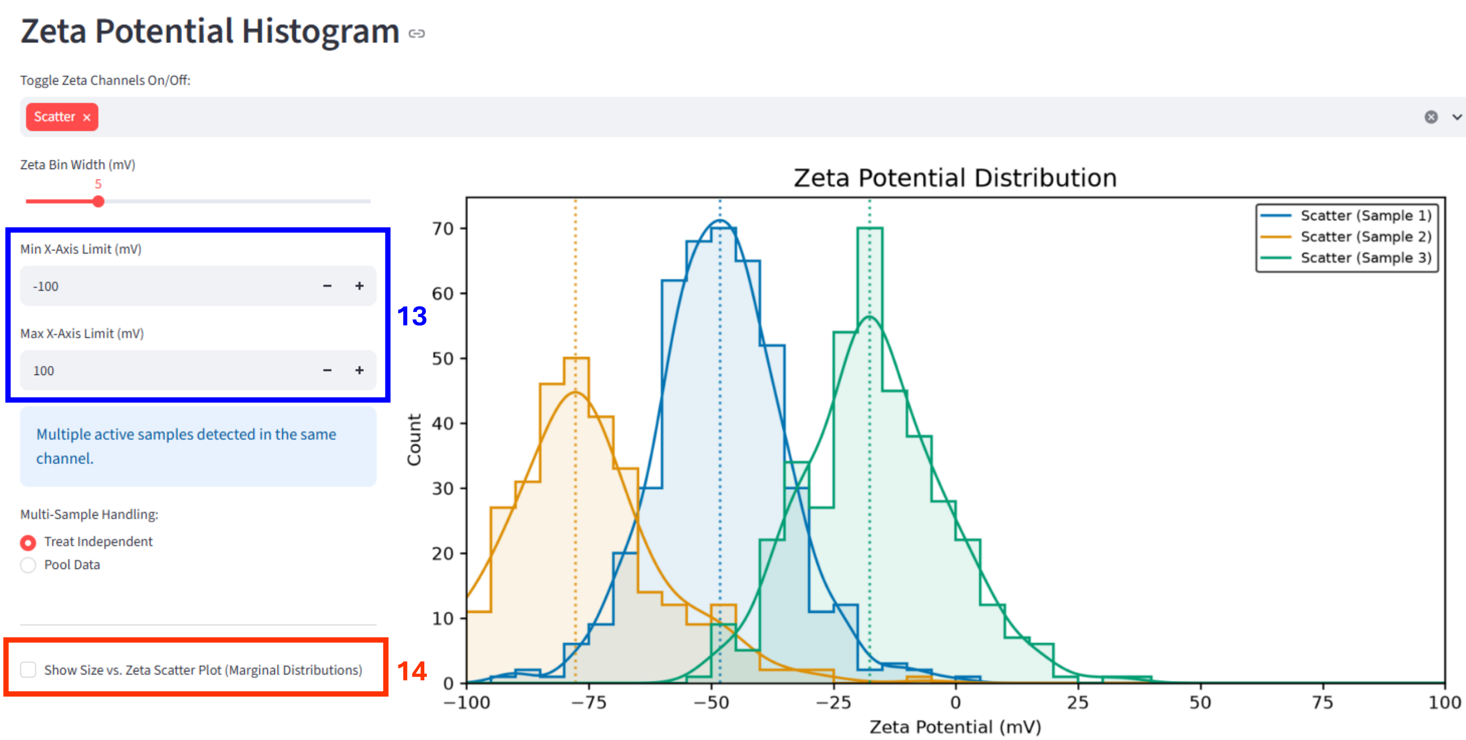
You can also use this tool to compare the scatter channel against any fluorescence channel. You can see in this page an example of a graph where scatter and the four fluorescence channels have been uploaded. The graph shows the BIC score for each channel and an overlap of all of them (with every fluorescence channel represented with the laser colour). Below, the sub-population graphs for each channel are represented, showing the different populations found on each of them (colours can be modified using the colour palette option).

When analysing scatter channel with one or more fluorescence channels, the global overview table will show extra data: **the positivity rate** (the ratio of particle counts in a fluorescent channel compared to the scatter channel. *Note: This is a bulk count comparison, not a measure of true single-particle co-localization*; and the statistics for the **comparison of every fluorescence channel against the scatter channel**.

There is a button below the graph to download an excel file containing all the data from the tables, the statistics, and all the data used to generate the population and BIC graphs, so you can replicate them in other software like Excel, GraphPad, etc.

Zeta Potential tab

Once your zeta potential file(s) are uploaded, you can visualize them in this tab.

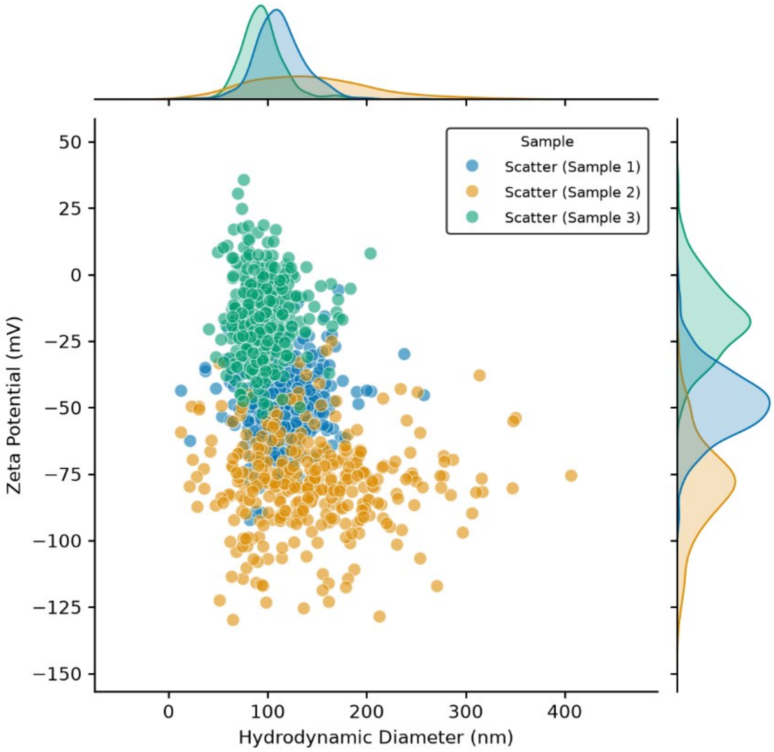


Zeta potential measurements are represented in mV, and the axis normally goes from -100 to +100, but the limits can be modified using the sliders **(13)**. Like in the Multi-Sample Size tab, the bin size can be modified and several samples can be pooled if they are replicates.

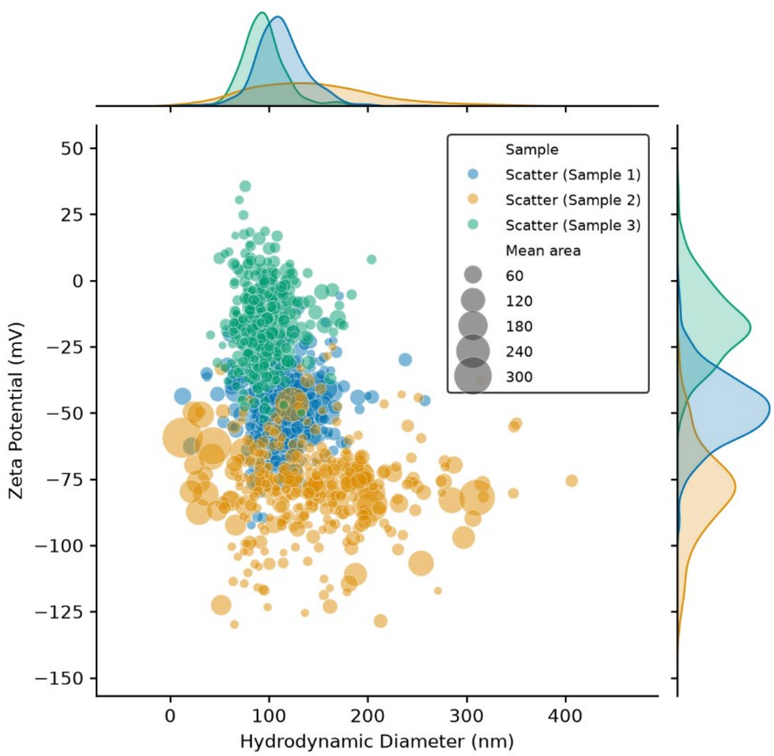
As the zeta potential raw data contains more information, including the size of the particles in that measurement, the zeta potential can be plotted against the particle size by clicking on **(14)**.

A new graph will display the size and zeta potential in one graph, and a new slider will allow to control the size of the dots.

A new option will also appear, allowing to show a 3rd variable as a bubble chart, where the size of the dots will represent the values of the 3rd variable (a list of variables will appear just above the combined graph).



The graph on the left represents the size vs zeta potential of three different samples.

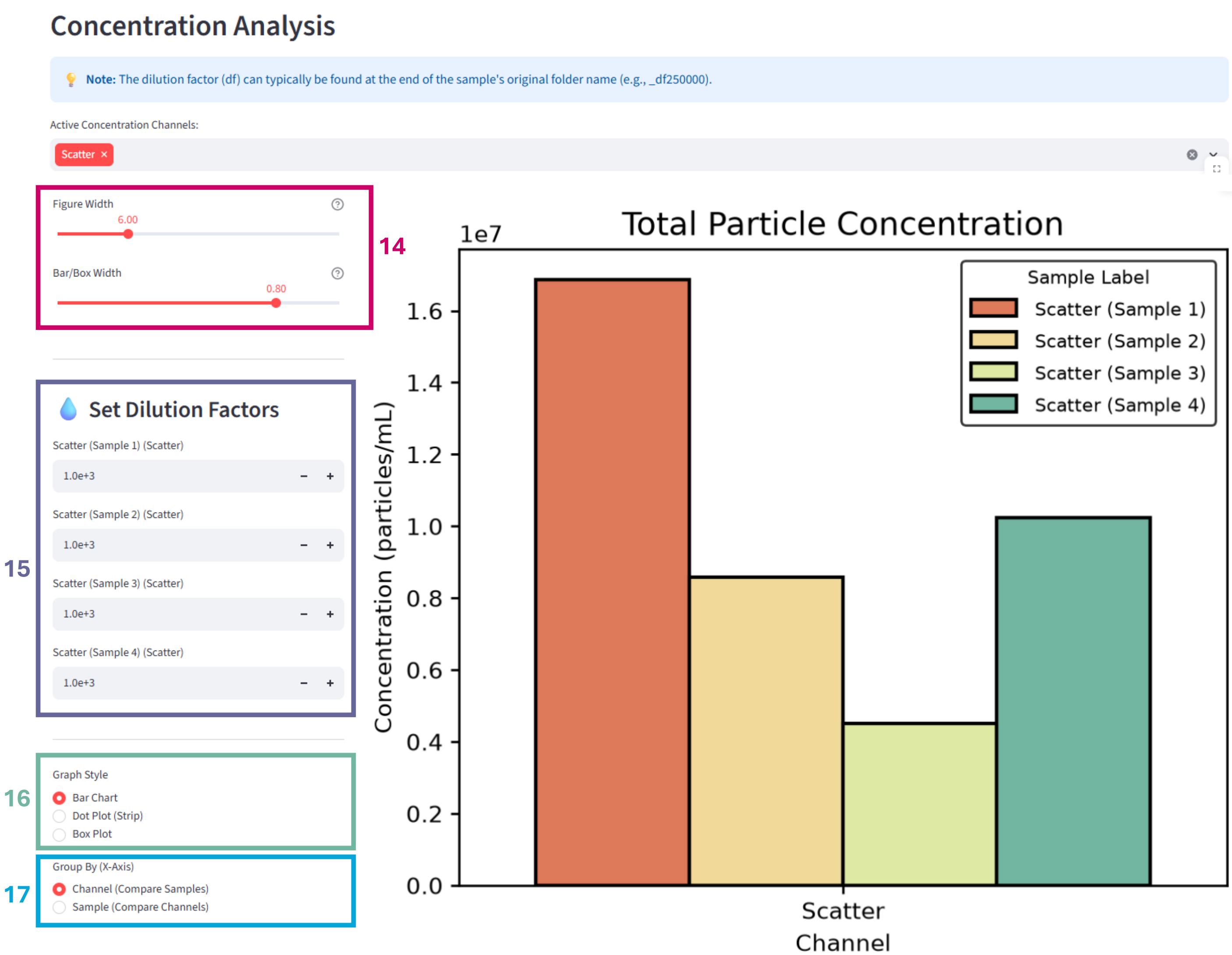


The graph on the left represents the size vs zeta potential of three different samples, and the size of the dots represents a 3rd variable, in this case the mean area.

Please take into account that the number of particles measured in a zeta potential measurement is much lower than the number of particles measured in a size measurement. Therefore, the size histograms on these combined graphs show less resolution and height and are never a substitute of a size measurement. Tables with count, mean, SD and statistics are also displayed (we work with mean, but not with median or mode with zeta potential results).

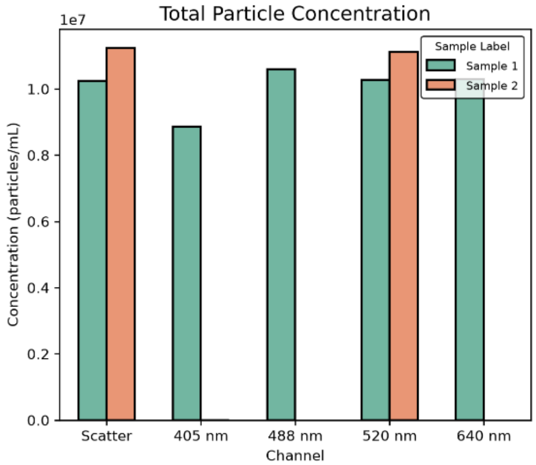
Concentration tab

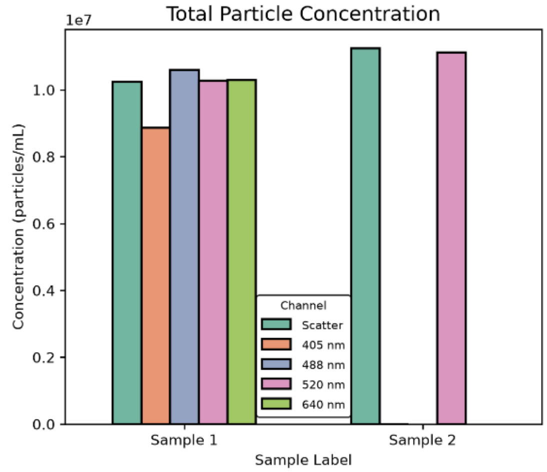
This section allows you to combine the concentration values for different samples/channels.



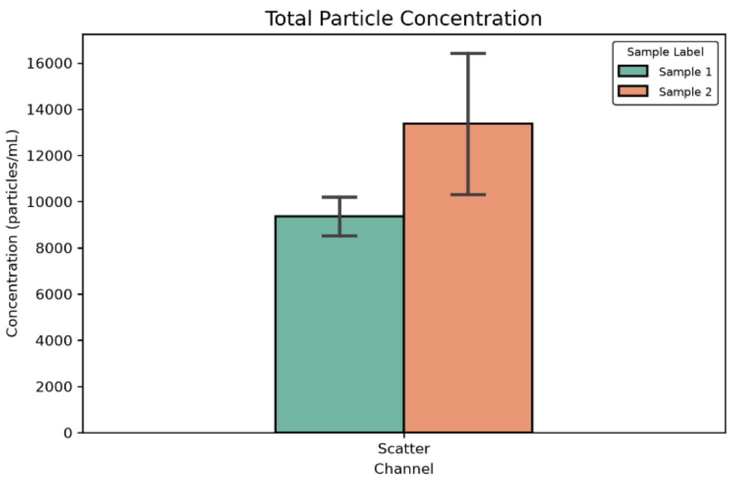
You can modify the width of the bars and the width of the graph on **(14)**. Unfortunately, the dilution factor of each sample must be entered manually on **(15)**, but you can find the dilution factor used in the name of the folder which contains you zip files (for example, df1000, at the end of the folder's name). By default, a bar graph is shown, but you can also select a dot plot or a box plot on **(16)**. You can switch to channel comparison (if different channels are uploaded) on **(17)**.

If you upload replicates, you can named them exactly with the same name on the "Customize Individual Samples" section, and the app will automatically group them and show the SD barrs.

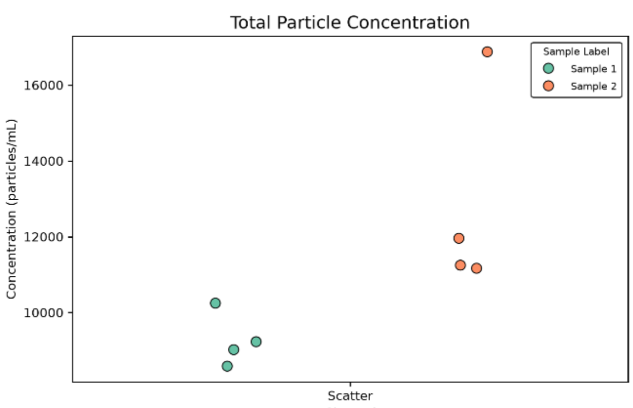


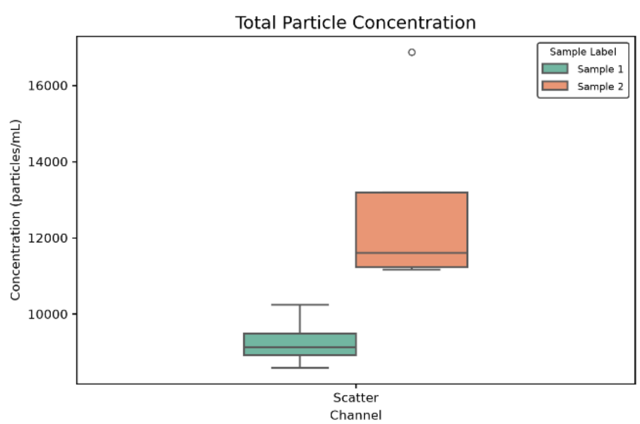


Example of comparing samples vs comparing channels.



When naming different replicates with the same name, they are grouped together and the SD bars are shown.





Examples of a dot plot and a box plot.

Colocalization tab

This tab will generate pie charts that show the percentage of colocalized particles (in yellow), and the particles that only positive for any of the two channels being compared. Just upload the colocalization zip files and the tab will automatically show a pie chart for every file uploaded.

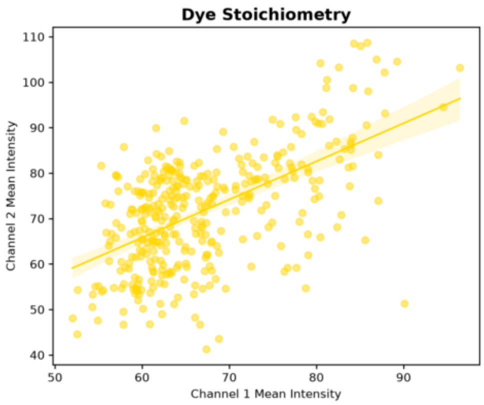
When visualizing more than one colocalization file, you can download the individual pie charts or a panel with all of them in a single image. A table with particle counts and the percentages is shown too. You can click on Override default colors with sidebar palette to use a colour palette from the list.

Colocalization Data Summary

	Sample Label	Only 405 nm Count	Only 488 nm Count	Colocalized Count	Total Unique Particles	Colocalized (%)	Only 520 nm Count	Only 640 nm Count
0	405 nm vs 488 nm (Sample 1)	93	217	367	677	54.2	None	None
1	405 nm vs 520 nm (Sample 1)	110	None	373	652	57.2	169	None
2	405 nm vs 640 nm (Sample 1)	92	None	227	393	57.8	None	74
3	488 nm vs 520 nm (Sample 1)	None	39	128	189	67.7	22	None
4	488 nm vs 640 nm (Sample 1)	None	53	137	214	64	None	24
5	520 nm vs 640 nm (Sample 1)	None	None	73	138	52.9	28	37

Colocalization Quality

This tab analyses the extra raw data in your colocalization files and calculates different quality parameters. There are for orientation purposes only, and must be interpreted with caution.



The Dye Stoichiometry plot is a scatter plot with a linear regression fit that compares the Mean Fluorescence Intensity of Channel 1 (x-axis) directly against Channel 2 (y-axis) for colocalized particle pairs only. Each dot represents a single detected particle with signal in both channels. The solid line represents the linear regression trend, showing the relationship between the dye signals across the sample population.

This graph can show if dual-labelled antibodies or probes possess uniform dye-to-protein labelling ratios. It can also indicate if the two target antigens, cargo molecules, or surface receptors are co-expressed in a consistent ratio.

The app calculates Pearson correlation (r) and coefficient of determination (R^2) and generates a table to score the results:

Stoichiometry Statistical Summary

Pearson Correlation (r)	R-squared (R^2)	Correlation Quality
0.575	0.331	Moderate (Medium)

Interpretation:

When $r \geq 0.70$ │ $R^2 \geq 0.49$ = Strong (Good)

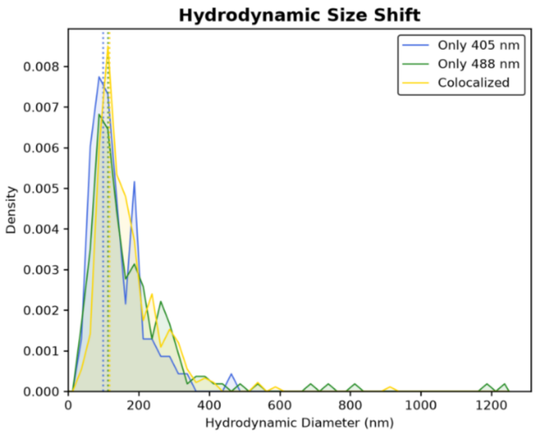
Proportional binding. As brightness increases in one channel, it increases proportionally in the other. High-confidence colocalization where particles carry a stable, uniform ratio of both markers.

When $0.40 \leq r \leq 0.70$ │ $0.16 \leq R^2 \leq 0.49$ = Moderate (Medium)

Partial correlation with significant dispersion. Particles carry both markers, but with variable stoichiometry. Common causes include heterogeneous receptor expression on biological vesicles or unequal fluorophore photobleaching.

When $r < 0.40$ │ $R^2 < 0.16$ = Weak (Poor)

Intensities are largely independent of one another. Dual-channel events are likely accidental spatial coincidences (e.g., high particle crowding where two independent particles fall within the link radius) rather than genuine single-particle complexes, or one fluorophore suffered severe degradation.



The Hidrodynamic Size Shift is a KDE plot that visualizes the distribution of physical particle diameters (in nm) for the single-positive vs colocalized populations.

The peaks of the curves show the most common physical size for each group. This plot helps you determine if the colocalized particles are physically distinct from the rest of the sample.

🟢 Consistent Size (Ratio < 1.2)

Dual-labelled particles share a nearly identical hydrodynamic size distribution with single-labelled particles. The colocalized population is physically similar to the bulk sample, and the dual-labelling protocol (e.g., adding two antibodies) did not severely alter their physical profile or induce aggregation.

🔴 Shift Detected (Ratio ≥ 1.2)

Colocalized particles have a notably larger hydrodynamic diameter (shifted to the right on the graph). Either your dual-labelling protocol is inducing physical aggregation in the sample, OR your biological markers are naturally and selectively expressed on only the largest vesicles/particles in your heterogeneous population.

2. Hydrodynamic Size Shift

405 nm Median Size	488 nm Median Size	Coloc. Median Size	Size Status
113.1 nm	134.0 nm	141.5 nm	🟢 Consistent Size

Interpretation: Dual-labeled particles share a nearly identical hydrodynamic size with single-labeled particles, confirming that dual-labeling does not severely alter their physical profile.

Tips

-You can directly drag & drop the folders for several samples. The app will load all the files contained in those folders, so all your measurements are loaded at once. The pdf report files will be detected as unknown, you can delete them from the list or just leave them.

-Click on the three dots in the upper-right corner to select among light mode, dark mode, or mimic your system mode.

-Place your pointer over a ⓘ symbol to see a helpful explanation.

-You can download any table by placing the pointer over it. A window will pop up to allow you download the table as a .csv file.

-You can visualize this help guide by clicking on the button at the bottom of the left side bar: